

Mathematical Operations

pow-math

Description: Calculates the power of a base raised to an exponent.

Parameters:

- Base: The base number.
- Power: The exponent.

Return: The result of the power function.

Example:

```
!(pow-math 2 3) ; Returns 8.0
```

sqrt-math

Description: Calculates the square root of a number.

Parameters:

- Number: The number to calculate the square root of. Must be ≥ 0 .

Return: The square root of the number.

Example:

```
!(sqrt-math 9) ; Returns 3.0
```

abs-math

Description: Calculates the absolute value of a number.

Parameters:

- Number: The number to calculate the absolute value of.

Return: The absolute value of the number.

Example:

```
!(abs-math -5) ; Returns 5
```

log-math

Description: Calculates the logarithm of a number with a given base.

Parameters:

- Base: The base of the logarithm.
- Number: The number to calculate the logarithm of.

Return: The result of the logarithm function.

Example:

```
!(log-math 10 100) ; Returns 2.0
```

trunc-math

Description: Returns the integer part of the input value

Parameters:

- Float: Input float value

Return: Integer part of float

Example:

```
!(trunc-math 5.6) ; Returns 5.0
```

ceil-math

Description: Returns the smallest integer greater than or equal to the input value

Parameters:

- Float: Input float value

Return: Integer value greater than or equal to the input

Example:

```
!(ceil-math 5.2) ; Returns 6.0
```

floor-math

Description: Returns the smallest integer less than or equal to the input value

Parameters:

- Float: Input float value

Return: Integer value less than or equal to the input

Example:

```
!(floor-math 5.8) ; Returns 5.0
```

round-math

Description: Returns the nearest integer to the input float value

Parameters:

- Float: Input float value

Return: Nearest integer to the input

Example:

```
!(round-math 5.4) ; Returns 5.0  
!(round-math 5.6) ; Returns 6.0
```

sin-math

Description: Returns result of the sine function for an input value in radians

Parameters:

- Angle: Angle in radians

Return: Result of the sine function

Example:

```
!(sin-math 0) ; Returns 0.0
```

asin-math

Description: Returns result of the arcsine function for an input value

Parameters:

- Float: Input float value

Return: Result of the arcsine function

Example:

```
!(asin-math 0) ; Returns 0.0
```

cos-math

Description: Returns result of the cosine function for an input value in radians

Parameters:

- Angle: Angle in radians

Return: Result of the cosine function

Example:

```
!(cos-math 0) ; Returns 1.0
```

acos-math

Description: Returns result of the arccosine function for an input value

Parameters:

- Float: Input float value

Return: Result of the arccosine function

Example:

```
!(acos-math 1) ; Returns 0.0
```

tan-math

Description: Returns result of the tangent function for an input value in radians

Parameters:

- Angle: Angle in radians

Return: Result of the tangent function

Example:

```
!(tan-math 0) ; Returns 0.0
```

atan-math

Description: Returns result of the arctangent function for an input value

Parameters:

- Float: Input float value

Return: Result of the arctangent function

Example:

```
!(atan-math 0) ; Returns 0.0
```

isnan-math

Description: Returns True if input value is NaN. False - otherwise

Parameters:

- Number: Number

Return: True/False

Example:

```
!(isnan-math 0.0) ; Returns False
```

isinf-math

Description: Returns True if input value is positive or negative infinity. False - otherwise

Parameters:

- Number: Number

Return: True/False

Example:

```
!(isinf-math 0.0) ; Returns False
```

min-atom

Description: Returns atom with min value in the expression. Only numbers allowed



latest



Parameters:

- Expression: Expression which contains atoms of Number type

Return: Min value in the expression. Error if expression contains non-numeric value or is empty

Example:

```
!(min-atom (2 6 7 4 9 3)) ; Returns 2.0
```

max-atom

Description: Returns atom with max value in the expression. Only numbers allowed

Parameters:

- Expression: Expression which contains atoms of Number type

Return: Max value in the expression. Error if expression contains non-numeric value or is empty

Example:

```
!(max-atom (2 6 7 4 9 3)) ; Returns 9.0
```

random-int

Description: Returns random int number from range defined by two numbers

Parameters:

- Range start: Range start
- Range end: Range end

Return: Random int number from defined range

Example:

```
!(random-int &rng 2 9) ; Returns any int number between 2 to 9
```

random-float

Description: Returns random float number from range defined by two numbers

Parameters:

- Range start: Range start
- Range end: Range end

Return: Random float number from defined range

Example:

```
!(random-float &rng 2 9) ; Returns any number in the interval [2, 9)
```